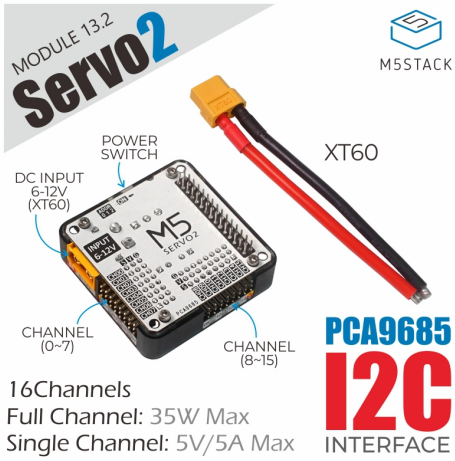
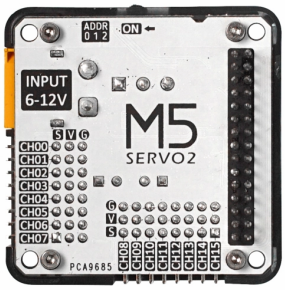
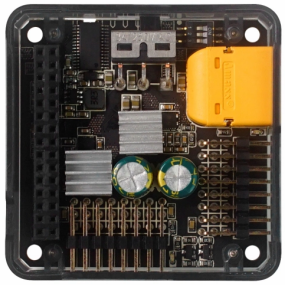
Module13.2 Servo2

SKU:M014-B





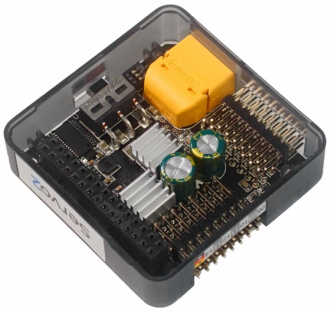












Description

Module13.2 Servo2 is a 16-channel servo driver module designed for M5Stack stackable controllers. It uses a PCA9685 12-bit PWM controller to independently control 16 servo signal channels and communicates with the host through M5-Bus I2C. The onboard 3-bit address DIP switch allows the I2C address to be configured from `0x40` (default) to `0x47`, supporting up to eight stacked modules (each module requires an independent power supply).

The logic circuitry and servo power supply are independent. The PCA9685 is powered by 3.3V from M5-Bus, while the 16 servo ports are divided into two groups of eight. Two SY8368AQQC step-down converters provide separate 5V power rails, with each converter powering one group of eight servo ports. With a DC 6 ~ 12V input through the XT60 connector, the two groups can simultaneously provide up to 5V/3.5A x 2 (35W total), while a single group can provide up to 5V/5A (25W). When powered by a compatible battery base, the maximum supported output is 5V/2A. The onboard power switch controls power to Module13.2 Servo2, and two green indicators show the status of the two 5V output groups.

Note

Module Power

The module must be externally powered through the DC connector when driving servos. For controllers with a power management IC (such as Core2/CoreS3), configure the M5-Bus power mode as input during program initialization (that is, `cfg.output_power = false`).

Features

- 16 x Servo Drive Channels
- 2 x Power Indicators
- Adjustable I2C Address
- Independent Power Switch Control
- External DC Power Input: 6-12V
- DC Connector Type: XT60

Includes

- 1 x Module13.2 Servo2
- 1 x XT60 Power Cable (11.5cm)

Applications

- Humanoid Robots
- Biomimetic Multi-Joint Robots
- 3-Axis Servo Gimbals

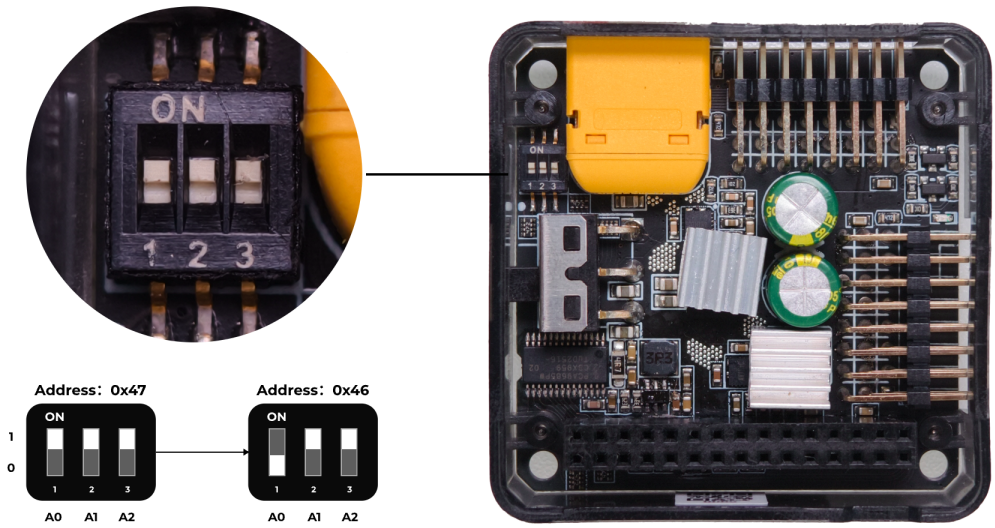
Specifications

Specification	Parameter
PWM Controller	PCA9685
PWM Resolution	12-bit (4096 steps)
Servo Channels	16 channels, 2 groups of 8
Communication Interface	I2C @0x40 (default) ~ 0x47
External Power Input	DC 6 ~ 12V
External Power Connector	XT60
Servo Output Voltage	5V
Dual-Group Load Capacity	5V/3.5A per group, 35W maximum total
Maximum Single-Group Load	5V/5A, 25W maximum
Battery Base Load Capacity	Up to 5V/2A
Power Indicators	2 x Green LEDs
Product Size	54.2 x 54.2 x 19.7mm
Product Weight	28.0g
Package Size	135.0 x 95.0 x 20.5mm
Gross Weight	60.0g

Learn

I2C Address Switching

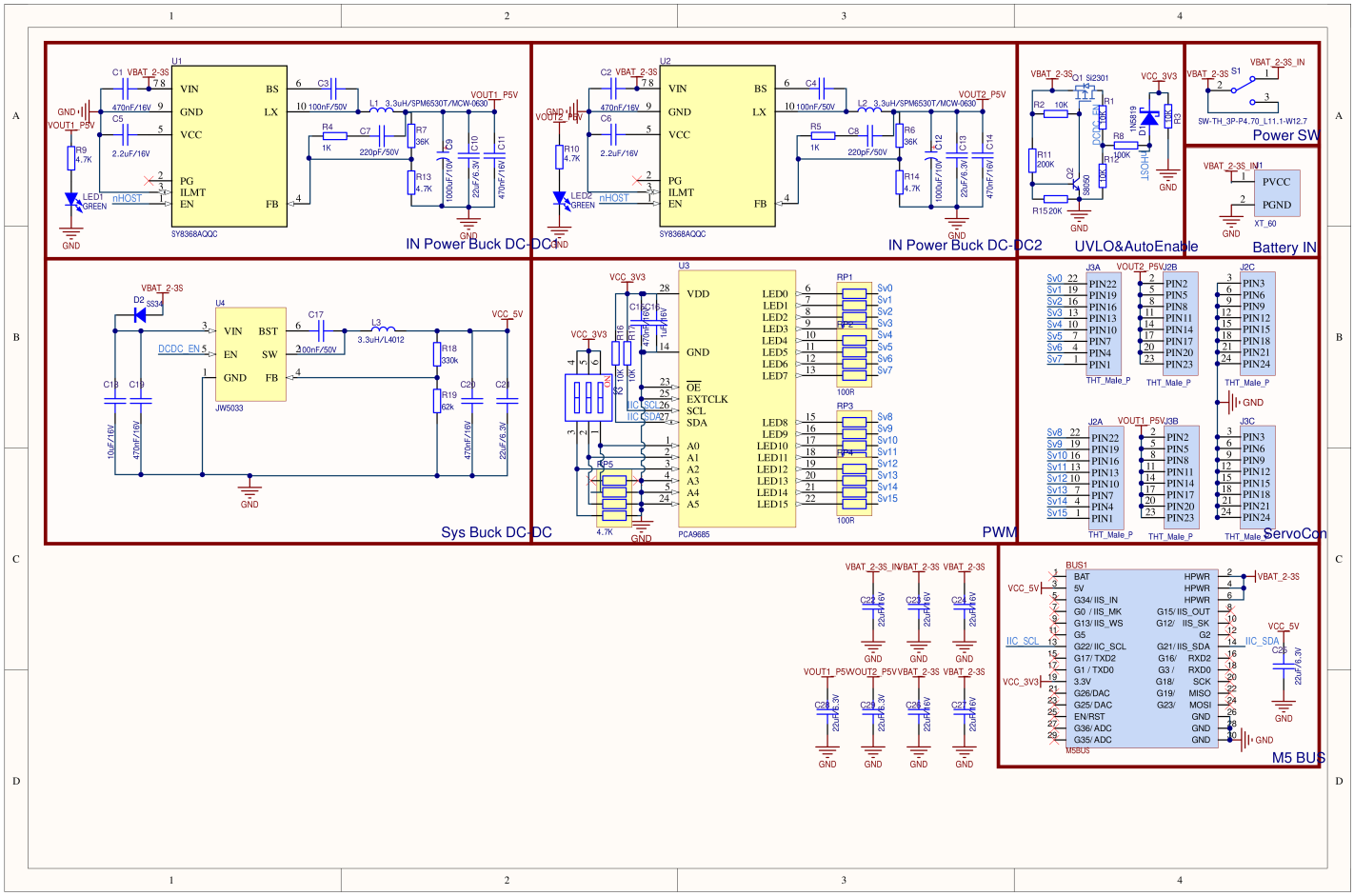
Module13.2 Servo2 has an onboard 3-bit DIP switch for configuring the I2C address. Switches 1, 2, and 3 correspond to address bits A0, A1, and A2, respectively. The UP position represents 1, and the DOWN position represents 0. Configure the DIP switch while the device is powered off, then power-cycle the device for the new address to take effect.



A0	A1	A2	7-bit I2C Address
0	0	0	0x40
1	0	0	0x41
0	1	0	0x42
1	1	0	0x43
0	0	1	0x44
1	0	1	0x45
0	1	1	0x46
1	1	1	0x47

Schematics

- Module13.2 Servo2 Schematics PDF



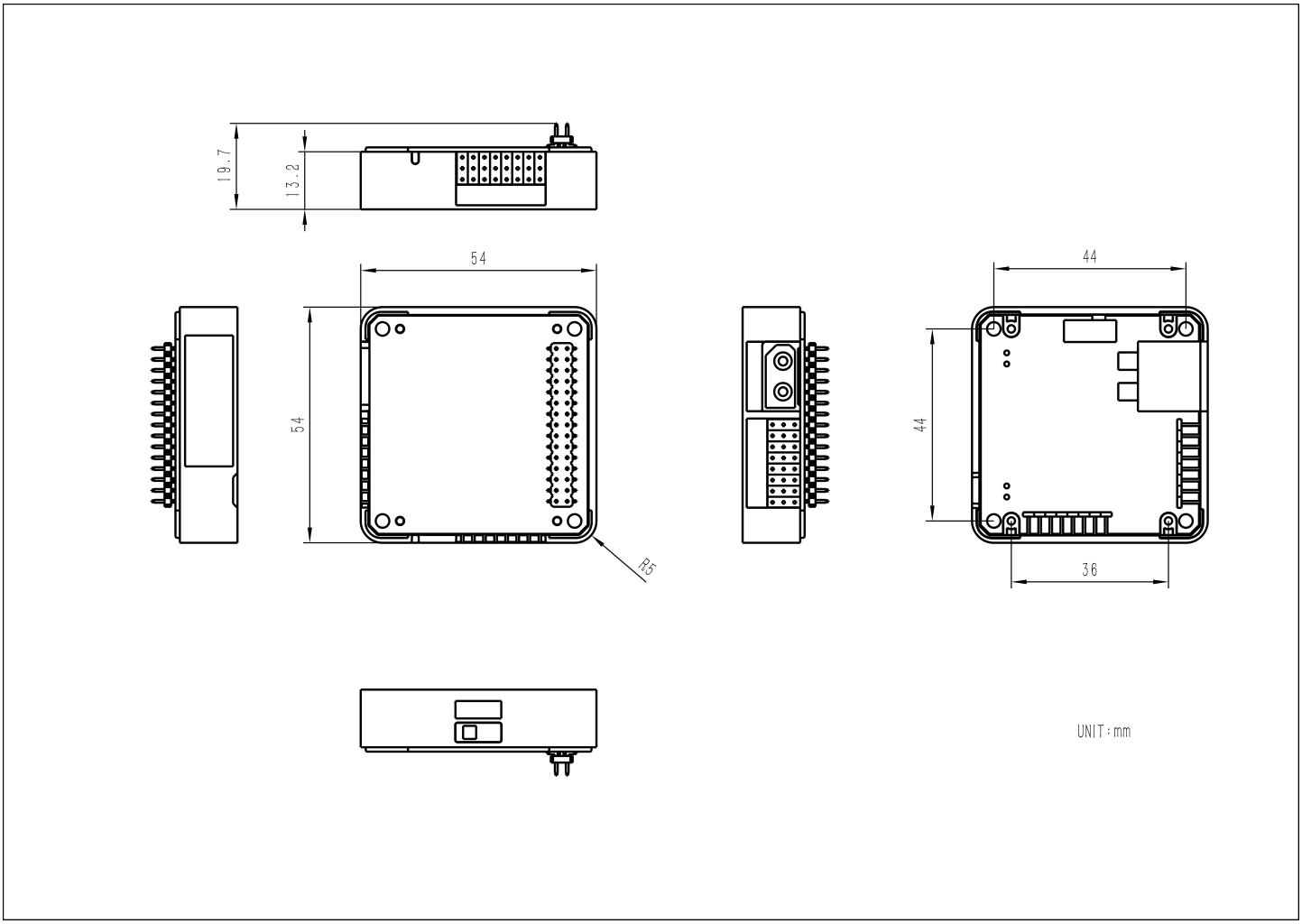
PinMap

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	
	7	8	
	9	10	
	11	12	3V3
	13	14	
	15	16	
SDA	17	18	SCL
	19	20	
	21	22	
	23	24	
HPWR	25	26	
HPWR	27	28	5V
HPWR	29	30	

Model Size

- Module13.2 Servo2 Model Size PDF



Datasheets

- PCA9685 - datasheet

Softwares

I2C Address Conflict

The onboard PCA9685 chip of Module13.2 Servo2 has the ALL_CALL broadcast address `0x70` enabled by default for synchronizing multiple chips. After power-on, the module responds not only to requests sent to its own I2C communication address (0x40 ~ 0x47), but also to the ALL_CALL I2C address `0x70`. If a device with the same address is connected to the current I2C bus, an address conflict will occur and communication may fail. This can be resolved by configuring the ALLCALL bit (bit:0) in the PCA9685 MODE1 register (0x00H) to `0` to disable the `ALL_CALL` function. See the Module13.2 Servo2 ALL_CALL Configuration example program.

Arduino

- Adafruit-PWM-Servo-Driver-Library
- Module13.2 Servo2 Arduino Tutorial

UiFlow1

- Module13.2 Servo2 UiFlow1 Docs

UiFlow2

- Module13.2 Servo2 UiFlow2 Docs

Video

Servo2.mp4